

Marketing Campaign EDA Summary Report

Objective

The objective of this project was to perform Exploratory Data Analysis (EDA) on the marketing campaign dataset to understand customer behavior, analyze purchasing patterns, and recommend where the marketing budget should be increased based on Return on Investment (ROI) and customer engagement.

Data Cleaning

- * Loaded the dataset into Python using Pandas.
- * Checked for missing values and found 24 missing values in the **Income** column.
- * Filled the missing values using the median income.
- * Checked for duplicate records and found none.
- * Converted the **Dt_Customer** column to datetime format.
- * Saved the cleaned dataset for further analysis.

Recommendations

Recommend where the company should spend more budget based on your charts. For example:

- * Increase investment in the **highest-performing purchase channel** (Web, Store, or Catalog).
- * Focus marketing campaigns on **high-income customers**, as they contribute more to sales.
- * Allocate more budget to promote **top-selling product categories** such as Wine (if it has the highest spending).
- * Design personalized campaigns for customers who did not respond to improve campaign effectiveness.
- * Continue analyzing customer behavior to optimize future marketing strategies.

Conclusion

The Exploratory Data Analysis helped identify customer purchasing behavior, product preferences, and effective sales channels. These insights can help the company make better marketing decisions and improve ROI by investing in the most successful products and channels.